Girijananda Chowdhury University

1st Sessioinal Exam (23/09/2024)

Course- B.Tech. [illegible] (CE)

Subject- Biology for Engineers

Subject Code-BBI23101T

Total marks=30

Duration-1hour

Q1. Answer all the MCQs (1X10=10)

a) Which of the following is not required in photosynthesis?
   i) Chlorophyll
   ii) Light
   iii) CO_2
   iv) O_2

b) Which of the following statements does not justify the importance of photosynthesis?
   i) It is the primary source of all food on earth.
   ii) It is responsible for the release of oxygen into the atmosphere.
   iii) It is responsibole for the water balance on earth.
   iv) Use of energy from sunlight by plants doing photosynthesis is the basis of life on earth.

c) In which color of wavelength, plants does the maximum absoorption of light?
   i) Green
   ii) Blue
   iii) Orange
   iv) Red

d) Photosynthetically active radiation (PAR) represents the following range of wavenlengths.
   i) 450-950 nm
   ii) 340-450 nm
   iii) 400-700 nm
   iv) 500-600 nm

e) The correct equation, representing the overall process of photosynthesis is
   i) $6\,CO_2 + 6\,H_2O \rightarrow C_6H_{12}O_6 + 6\,H_2O + 6\,O_2$
   ii) $6\,CO_2 + 6\,H_2O \rightarrow C_6H_{12}O_6 + 6\,O_2$
   iii) $6\,CO_2 + 6\,H_2O \rightarrow C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$
   iv) $6\,CO_2 + 6\,H_2O \rightarrow C_6H_{12}O_6 + 6\,O_2 + H_2O$

f) Glycolysis is found in cytoplasm of virtually all types of aerobic/ anaerobic cells. In this process, glucose is vonverted into:
   i) PEP
   ii) Acetyl CoA
   iii) Pyruvic acid
   iv) Citric acid

g) The energy yielding steps of glycolysis are:

i) Conversion of PGA to BPGA and conversion of PGAL to DHAP.
ii) Conversion of PGAL to PGA and conversion of BPGA to PGA.
iii) Conversion of BPGA to PGA and conversion of PEP to Pyruvate
v) Conversion of pyruvate to PEP and PGAL to DHAP

h) What is the total amount of ATP produced during Citric acid Cycle/ Krebs cycle?

i) 4
ii) 2
iii) 6
v) 8

i) Glycolysis is :

i) Oxidation of Glucose to Glutamate
ii) Conversion of Pyruvate to Citrate
iii) Oxidation of Glucose to Pyruvate
v) Conversion of Glucose to Haem

j) Product of Kreb cycle essential for oxidative phosphorylation is:

i) NADPH and ATP
ii) Acetyl CoA
iii) CO_2 and Oxaloacetate
iv) NADH and $FADH_2$

Q2. Answer any four Questions (5X4=20)

a) What is Photosynthesis? Where does it occur in a plant, show in a diagram? Mention the stages invoved in it and their respective place of occurence? $(1_{1/2}+1_{1/2}+2=5)$

b) What is light reaction? Explain with a suitable diagram. (1+4=5)

c) What is dark reacction? Mention the important steps in volved in it. (1+4=5)

d) Write in brief the summary of cellular respiration. 5

e) *What are the proteing complexes* invoved in oxidative phosphirylation? Write the mechanism behind synthesis *of ATP.* (2+3=5)